BOREAS HYD-04 Areal Snow Course Data

Summary

The BOREAS HYD-04 work was focused on collecting data during the winter field campaign (FFC-W) to improve the understanding of winter processes within the boreal forest. Knowledge of snow cover and its variability in the boreal forest is fundamental if BOREAS is to achieve its goals of understanding the processes and states involved in the exchange of energy and water. The development and validation of remote sensing algorithms will provide the means to extend the knowledge of these processes and states from the local to the regional scale. A specific thrust of the hydrology research is the development and validation of snow cover algorithms from airborne passive microwave measurements.

Airborne remote sensing data (gamma, passive microwave) were acquired along a series of flight lines established in the vicinity of the BOREAS study areas. Ground snow surveys were conducted along selected sections of these aircraft flight lines. These calibration segments were typically 10-20 km in length, and ground data were collected at 1- to 2-km intervals. The data are provided in tabular American Standard Code for Information Interchange (ASCII) files.

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1. Data Set Overview

1.1 Data Set Identification

BOREAS HYD-04 Areal Snow Course Data

1.2 Data Set Introduction

These ground snow survey data were collected to calibrate gamma and passive microwave remote sensing data that were acquired in the BOREAS study areas.

These data were collected during the BOREAS Focused Field Campaign-Winter (FFC-W) in 1994 and cover 10-20 km lengths with measurements representing the snow conditions for 1- to 2-km intervals along those lengths. This data set should be used in conjunction with the HYD-06 Airborne Gamma Ray Soil Moisture Data since these 2 data sets cover the same sites. Please consult the documentation for this HYD-06 data set for more information about the transects.

1.3 Objective/Purpose

These data were collected during the BOReal Ecosystem Atmosphere Study (BOREAS) FFC-W, which took place during the period 02-18-Feb-1994. These data were collected to provide detailed spatial snow depth and snow water equivalent (SWE) measurements in the BOREAS study areas to validate airborne remote sensing techniques and to serve as the basis in algorithm development to determine SWE using passive microwave data. These data will also provide background information for other BOREAS hydrological and satellite remote sensing components.

1.4 Summary of Parameters

Flight line number
location (latitude/longitude)
average snow water
SWE
depth and density
number of samples
standard deviation of SWE and depth
general land cover characteristics

1.5 Discussion

Ground snow survey data were collected along preselected calibration lines of the airborne snow survey flight line network. Along each calibration line, snowpack information was obtained at a series of equally spaced locations along the line (e.g., 1 or 2 km). A concerted effort was made to locate sampling points in vegetation of the type and approximate percentage cover representative of the calibration flight line. Up to three SWE and depth measurements plus 15 additional snow depth measurements were taken at each sampling location.

Data were collected along 15 calibration lines in the Southern Study Area (SSA), 3 lines in the Northern Study Area (NSA) and 5 lines along a transect between the two study areas. Each calibration line was subsequently subdivided into approximately equal length segments called bins. Each bin represents an area of similar vegetation, determined by detailed land cover analysis of the BOREAS land classification by HYD-04, which was then provided to HYD-06 for processing the airborne gamma data. All reference coordinates are taken from Global Positioning System (GPS) North American Datum of 1983 (NAD83) sampling point locations averaged to represent the approximate center of the calibration line or bin.

1.6 Related Data Sets

BOREAS HYD-04 Standard Snow Course Data

BOREAS HYD-06 Aircraft Gamma Ray Soil Measurement Data
BOREAS HYD-06 Ground Measurements of Soil Moisture
BOREAS HYD-06 Moss/Humus Moisture Data
BOREAS HYD-03 Snow Measurements

2. Investigator(s)

2.1 Investigator(s) Name and Title

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2.2 Title of Investigation

Determination of snow cover variations in the boreal forest using passive microwave radiometry.

2.3 Contact Information

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3. Theory of Measurements

Snow sampling equipment consists of a graduated tube with a cutter fixed to its lower end to permit easy penetration of the snow and a spring balance (reading directly in water equivalent units) to weigh the tube and its contents. The density of the snow is calculated by dividing the SWE by the depth of the snow at the sample point. Additional snow depth measurements were made with a graduated stick or ruler.

4. Equipment

4.1 Sensor/Instrument Description

The large diameter Eastern Snow Conference (ESC)-30 metric snow sampler with a cutter area of 30 cm^2 was used to obtain SWE measurements. A complete description of this sampler along with plans, specifications, and an assessment of errors and accuracy can be found in Farnes et al., 1982. The ESC-30 sampler is the current standard sampler used by the Atmospheric Environment Service (AES) for shallow snowpack regions. The following is a complete list of equipment supplied to the surveyors:

1 ESC-30 snow sampler, 1 spring balance for ESC-30, 1 cradle, 1 measuring stick/ruler (cm), field book or snow survey form and pencil.

4.1.1 Collection Environment

These data were collected under snow cover conditions at various sites in the SSA and NSA.

4.1.2 Source/Platform

Measurements were made at selected sites, predetermined from a fixed point at the start of the calibration line. Individual points were selected randomly in a circle extending approximately 50 m in from the edge of the forest. Ground irregularities (e.g., boulders, logs) were avoided.

4.1.3 Source/Platform Mission Objectives

The objective of collecting this data was to provide ground-based snow measurements coincident with airborne microwave and gamma airborne missions for algorithm development and validation.

4.1.4 Key Variables

Snow depth, density, water equivalent, land cover characteristic, and snowpack structure.

4.1.5 Principles of Operation

The snow sampler is lowered vertically into the snowpack with a steady thrust downward. A small amount of twisting aids in driving the tube and cutting thin ice layers; however, considerable force and driving of the sampler may be required to penetrate hard layers of ground ice. Penetration to extract a soil plug helps to prevent the loss of the snow core from the tube; a trace of soil or litter in the cutter indicates that no loss has occurred. Observation of the

length of the snow core permits a quick assessment of whether a complete core is obtained; the depth of snow is measured when the sampler is inserted in the snowpack. The sample is weighed in the tube and the combined weight (in water equivalent units) is read directly with the spring balance. The tare weight of the tube is subtracted to obtain the SWE. The density of the snow is determined by dividing the water equivalent by the depth of the snow. A graduated meter stick is used in obtaining ancillary snow depth measurements.

More detailed information on suggested snow survey procedures is available in Snow survey and water supply forecasting (U.S. Soil Conservation Service, 1972), Snow surveying (Atmospheric Environment Service, 1973), and Guide to Hydrological Practices (World Meteorological Organization, 1974).

4.1.6 Sensor/Instrument Measurement Geometry

The ESC snow sampler extracts a snow core with a surface area of 30 cm^2.

4.1.7 Manufacturer of Sensor/Instrument

The ESC-30 snow sampler and spring balance are manufactured and calibrated according to Atmospheric Environment Service (AES) specifications. The ESC-30 snow sampler was made under contract for AES. There are no commercial manufacturers of this instrument. Please see specifications on how to make the snow tube in Farnes et al., 1982.

4.2 Calibration

The spring balance is used to weigh snow samples. The balance is held in a free position, and the snow sampler tube containing the snow sample is suspended from the balance in a special cradle. The inner and outer tubes of the balance are made of anodized aluminum. The anchorages for the spring in the outer and inner tubes are designed to permit free, unrestricted flexing of the spring over its entire effective length. The balance has a ring at the top for hanging the balance and a hook at the other end from which the snow sampler cradle is hung. The inner tube has a scale engraved into it, which reads from 0 at the bottom to 125 at the top. The scale indicates the SWE in centimeters.

4.2.1 Specifications

Calibration is performed by the manufacturer using a set of known weights to adjust the magnification of the spring. For more information, see AES spring balance calibration procedure, February 1987, issue 2.

4.2.1.1 Tolerance

A balance must read within ±0.3 cm of the actual weight at all levels up to 60.5 cm and ±0.4 cm at levels above 60.5 cm.

4.2.2 Frequency of Calibration

Spring balances are checked annually before snow surveys begin. A pail and water (1 liter) can be used to check the accuracy of the balance. First, use a pail of water to register a positive reading on the balance, then add 1 liter

of water to the pail. The difference between the two scale readings should equal 37.8 ±0.3 cm.

4.2.3 Other Calibration Information

The spring balance should be checked from time to time during the season. Consistent repeat readings (i.e., tare) should be observed, with no indication of appreciable sticking of the spring in the balance. This ensures that the balance is not giving a false reading because the spring is not moving freely.

5. Data Acquisition Methods

Procedures similar to those described in the data set identified as Standard Snow Surveys were used. However, 3 SWE and 15 additional depth measurements had to be made at each sampling location. The individual sampling points were randomly selected (not predetermined) and were marked with a T-bar fence post.

6. Observations

Observations include detailed snowpack statigraphy, including ground, snow, snow surface temperatures, and snow grain type and size.

6.1 Data Notes

None given.

6.2 Field Notes

Copies of field notes are available from the primary contact.

7. Data Description

7.1 Spatial Characteristics

7.1.1 Spatial Coverage

These data were collected at the two BOREAS study areas in coordination with aircraft measurements made by the BOREAS HYD-06 team.. The two study areas are located within a large area of interest covering over a million square kilometers in the Canadian Provinces of Saskatchewan and Manitoba. Each of the study areas is approximately 50 by 100 km.

The HYD-06 aircraft flew at an altitude of 150 m AGL and measured natural terrestrial gamma radiation, on average, over an average path 10.7 km long and 305 m wide, or an areal extent of 3.3 km2. Water content was collected at various points along the flight. The number and timing of the flight lines flown each day were determined by the need for ground calibration and on the basis of need for other studies (priority was given where possible to obtaining measurements over flight lines associated with ground tower sites) and the amount of flight hours remaining for the aircraft before required maintenance.

These measurements were made within the NSA and SSA and along a transect that is between these two study areas.

There is a reference file associated with the HYD-06 data that contains information about the location of the various flight lines.

NSA Spatial Coverage (North American Datum 1983 (NAD83))

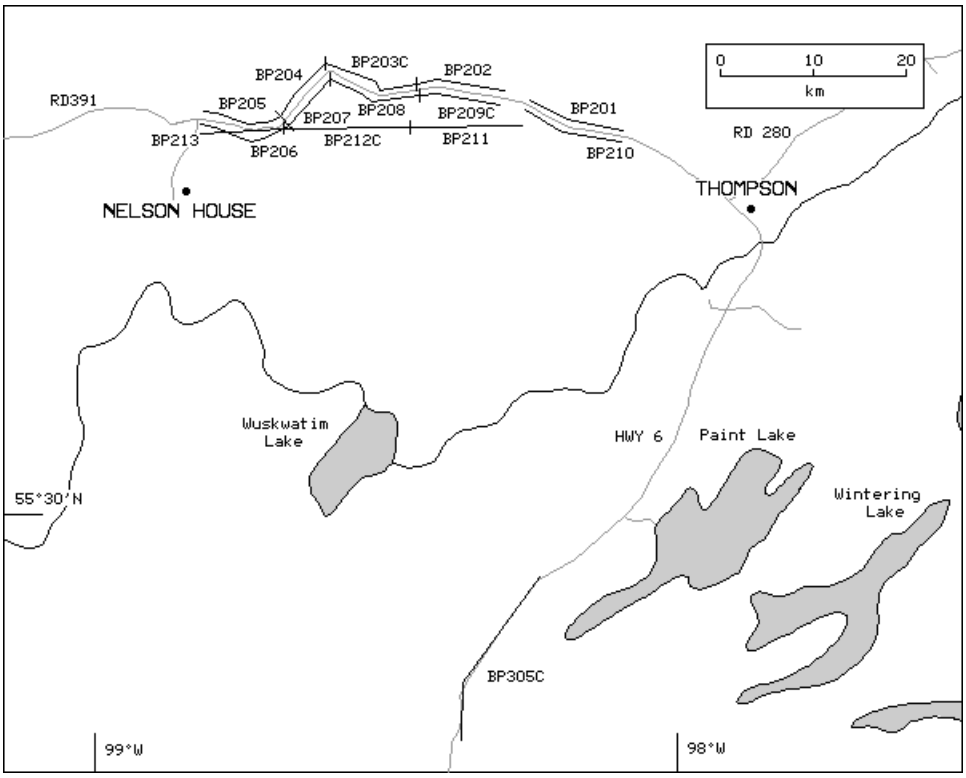
	Longitude	Latitude
Upper Left	-98.82	56.247
Upper Right	-97.24	56.081
Lower Right	-97.49	55.377
Lower Left	-99.05	55.54

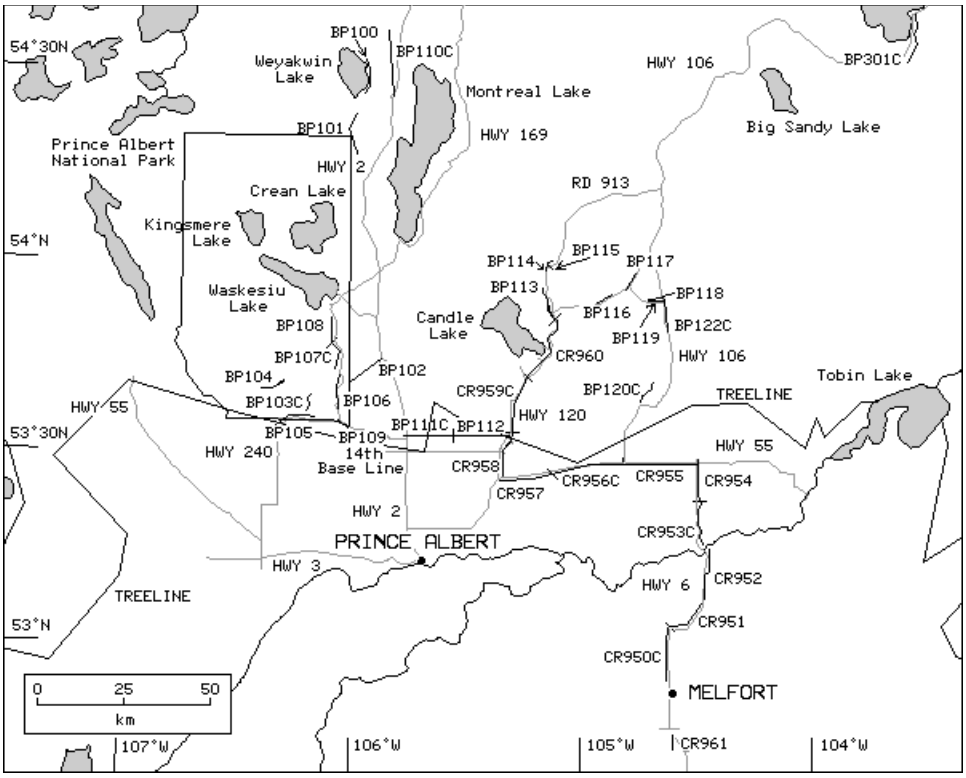
SSA Spatial Coverage

	Longitude	Latitude
Upper Left	-106.23	54.319
Upper Right	-104.24	54.223
Lower Right	-104.37	53.419
Lower Left	-106.32	53.513

7.1.2 Spatial Coverage Map

Maps of the study sites are given in the following diagrams.





7.1.3 Spatial Resolution

These data are point source measurements at the locations indicated.

7.1.4 Projection

Not applicable.

7.1.5 Grid Description

Not applicable.

7.2 Temporal Characteristics

7.2.1 Temporal Coverage

These data were collected from 03-Feb-1994 to 10-Mar-1994.

7.2.2 Temporal Coverage Map

The list of sites and dates when the data were collected is too long to list here. Most measurements took place between 04-FEB-1994 and 11-FEB-1994. Data for flight line BP212 was collected on 10-MAR-1994.

7.2.3 Temporal Resolution

Data were collected at a single point in time for the various sites.

7.3 Data Characteristics

Data characteristics are defined in the companion data definition file (h04assd.def).

7.4 Sample Data Record

Sample data format shown in the companion data definition file (h04assd.def).

8. Data Organization

8.1 Data Granularity

All of the Areal Snow Course Data are contained in one dataset.

8.2 Data Format(s)

The data files contain numerical and character fields of varying length separated by commas. The character fields are enclosed with a single apostrophe marks. There are no spaces between the fields. Sample data records are shown in the companion data definition files (h04assd.def).

9. Data Manipulations

9.1 Formulae

9.1.1 Derivation Techniques and Algorithms

density = (swe/(depth*10))*1000

9.2 Data Processing Sequence

9.2.1 Processing Steps

None given.

9.2.2 Processing Changes

None.

9.3 Calculations

9.3.1 Special Corrections/Adjustments

None.

9.3.2 Calculated Variables

None.

9.4 Graphs and Plots

None.

10. Errors

10.1 Sources of Error

Errors in the data collected by the snow survey teams may arise from several sources:

1) Warm temperatures, which can cause snow to stick in the tube.
2) Changes to scale precision caused by air temperature fluctuations.
3) Scale readings by different observers or during windy conditions, which can yield erratic scale readings.

10.2 Quality Assessment

The quality of the data collected by the survey teams is thought to range from good to excellent; generally it is considered to be excellent. Generally, data quality is potentially lowest when the snowpack is shallowest. No formal quality assurance is done; however, a coarse check was made after the completion of the survey.

10.2.1 Data Validation by Source

The data were spot checked by BORIS to make sure that no conversion errors occurred during loading.

10.2.2 Confidence Level/Accuracy Judgment

This data set represents the best estimate of snow depth and SWE by land cover/vegetation type for the immediate vicinity of the snow course on the date of the survey. Care should be exercised when extrapolating the measurements in time and/or to a much larger geographic area.

10.2.3 Measurement Error for Parameters

Given by standard error for SWE and depth.

10.2.4 Additional Quality Assessments

None.

10.2.5 Data Verification by Data Center

The data set was reviewed to ensure that it represents the data that are described.

11. Notes

11.1 Limitations of the Data

This data set represents the best estimate of snow depth and SWE by land cover/vegetation type for the immediate vicinity of the snow course on the date of the survey. Care should be exercised when extrapolating the measurements in time and/or to a much larger geographic area.

11.2 Known Problems with the Data

None given.

11.3 Usage Guidance

Care should be exercised when extrapolating the measurements in time and/or to a much larger geographic area.

11.4 Other Relevant Information

None.

12. Application of the Data Set

This data set can be used to develop and validate snow cover algorithms from airborne passive microwave measurements or other remote sensing techniques where SWE can be estimated.

13. Future Modifications and Plans

None.

14. Software

14.1 Software Description

None.

14.2 Software Access

None.

15. Data Access

15.1 Contact Information

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15.2 Data Center Identification

See Section 15.1.

15.3 Procedures for Obtaining Data

Users may place requests by telephone, electronic mail, or fax.

15.4 Data Center Status/Plans

The BOREAS HYD-04 areal snow course data are available from the Earth Observing System Data and Information System (EOSDIS) Oak Ridge National Laboratory (ORNL) Distributed Active Archive Center (DAAC).

The BOREAS contact at ORNL is:

ORNL DAAC User Services
Oak Ridge National Laboratory
(865) 241-3952
ornldaac@ornl.gov
ornl@eos.nasa.gov

16. Output Products and Availability

16.1 Tape Products

None.

16.2 Film Products

None.

16.3 Other Products

The data and documentation are available as ASCII files.

17. References

17.1 Platform/Sensor/Instrument/Data Processing Documentation

Atmospheric Environment Service. 1973. Snow surveying. 2nd Ed. Environ. Can., Downsview. Ont.

Farnes, P.E., N.R. Peterson, B.E. Goodison and R.P. Richards. 1982. Metrication of Manual Snow Sampling Equipment by Western Snow Conference Metrication Committee. 39th Annual Proceedings of the Eastern Snow Conference, Reno, NV, Apr. 19-23, 1982, pp. 120-132.

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U.S. Soil Conservation Service. 1972. Snow survey and water supply forecasting. Section 22, SCS Nat. Eng. Handb., U.S. Dept. Agric., Washington, DC.

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17.2 Journal Articles and Study Reports

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(CD-ROM) GERí97 (Geomatics in the Era of Radarsat). Ottawa, Canada, May 25-30, 1997.

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Walker, A.E., B.E. Goodison, and J.R. Metcalfe. 1995. Investigation of Boreal Forest Snow Cover Variations During the BOREAS 1994 Winter Campaign. 17th Canadian Symposium on Remote Sensing, Saskatoon, Saskatchewan, June 13-15, 1995.

Walker, A.E., B.E. Goodison, and J.R. Metcalfe. 1995. Investigation of Boreal Forest Snow Cover Variations During the BOREAS 1994 Winter Campaign. 29th Canadian Meteorological and Oceanographic Society Congress, Kelowna, British Columbia, May 30-June 2, 1995.

17.3 Archive/DBMS Usage Documentation

None.

18. Glossary of Terms

None.

19. List of Acronyms

AES	- Atmospheric Environment Service
ASCII	- American Standard Code for Information Interchange
BOREAS	- BOReal Ecosystem-Atmosphere Study
BORIS	- BOREAS Information System
CGR	- Certified by Group
CPI	- Certified by Principal Investigator
CPI-???	- CPI but questionable
DAAC	- Distributed Active Archive Center
EOS	- Earth Observing System

EOSDIS - EOS Data and Information System
ESC - Eastern Snow Conference
FFC-W - Focused Field Campaign - Winter
GSFC - Goddard Space Flight Center
HYD - Hydrology
NASA - National Aeronautics and Space Administration
NSA - Northern Study Area
ORNL - Oak Ridge National Laboratory
PANP - Prince Albert National Park
PI - Principal Investigator
PRE - Preliminary
SSA - Southern Study Area
SWE - Snow Water Equivalent
URL - Uniform Resource Locator

20. Document Information

20.1 Document Revision Date

Written: 01-May-1995
Last Revised: 06-Jul-1998

20.2 Document Review Date

BORIS Review: 26-Feb-1998
Science Review: 13-Jan-1998

20.3 Document ID

20.4 Citation

The areal snow survey data were collected by personnel with the Climate Research Branch, AES, with assistance from personnel from the University of Waterloo, Dept. of Civil Engineering, NASA GSFC, and the Norwegian Institute of Technology.

20.5 Document Curator

20.6 Document URL

Keywords

SNOW DENSITY
SNOW WATER EQUIVALENT